

Section : Biology

Q.1 Which one of the following is not the invasive alien species?

1. *Lantana*
2. Thylacine
3. *Clarias gariepinus*
4. *Parthenium*

Options 1. 1

2. 2
3. 3
4. 4

Q.2 Which one of the following is a correct statement about dormancy?

1. Actively respiring embryo in a seed.
2. Embryonal state of inactivity.
3. Embryo in early stage of development.
4. Presence of one embryo in a seed.

Options 1. 1

2. 2
3. 3
4. 4

Q.3 Which one of the following statements is not correct regarding the DNA?

1. DNA as an acidic substance present in nucleus was first identified by Friedrich Meischer in 1869.
2. The two chains are coiled in a left-handed fashion and the pitch of the helix is 3.4 nm.
3. The plane of one base pair stacks over the other in double helix and confers stability of the helical structure.
4. The two chains have anti-parallel polarity, means if one chain has the polarity 5' to 3', the other has 3' to 5'.

Options 1. 1

2. 2
3. 3
4. 4

Q.4 Amniocentesis is not legally used to test:

1. Haemophilia
2. Sex of the foetus
3. Thalassaemia
4. Sickle cell anemia

Options 1. 1

2. 2
3. 3
4. 4

Q.5 Match List-I with List-II

List-I

List-II

Terms

Features

- | | |
|-----------------|---------------------------|
| (A) Natality | (I) Exit of individuals |
| (B) Mortality | (II) Entry of individuals |
| (C) Immigration | (III) Death rate |
| (D) Emigration | (IV) Birth rate |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
3. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
4. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

Options 1. 1

2. 2
3. 3
4. 4

Q.6 Arrange the given products formed during sewage treatment in correct sequence?

- (A) Biogas
- (B) Activated sludge
- (C) Floccs
- (D) Primary sludge

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (D), (C), (B), (A)
3. (C), (D), (B), (A)
4. (B), (A), (D), (C)

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Match List-I with List-II

Terms

Examples

- | | |
|--------------------------|-----------------------------------|
| (A) Somatic hybrid | (I) Bt toxin |
| (B) Tissue culture | (II) Apical and axillary meristem |
| (C) Insecticidal protein | (III) Pomato |
| (D) Virus free plant | (IV) Explants |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.8 Which one of the followings can helps in destroying the tumor by activating the immune system of the patient?

1. Opioids
2. β -interferon
3. α -interferon
4. Cannabinoids

Options 1. 1

2. 2
3. 3
4. 4

Q.9 Which one of the following is not fermented by microbes to make foods?

1. Fish
2. Soyabean
3. Rock salt
4. Bamboo shoots

Options 1. 1

2. 2
3. 3
4. 4

Q.10 Which one of the following takes place in all eukaryotic organisms as a method of cellular defense?

1. dsRNA
2. mRNA

3. RNAi

4. DNA

Options 1. 1

2. 2

3. 3

4. 4

Q.11 After industrialization in 1920 in England, the proportion of dark winged and white winged moths reversed because of:

1. The tree trunks became light in color.
2. The tree trunks became dark due to soot.
3. The moths' wings became dark due to soot and smoke.
4. Dark colored moths were easily spotted by predators.

Options 1. 1

2. 2

3. 3

4. 4

Q.12 ZIFT stands for:

1. Zygote Intra Fertilization Technique
2. Zygote Intra Fallopian Transfer
3. Zygote Insemination Fertile Technique
4. Zygote Inter Foetus Transfer

Options 1. 1

2. 2

3. 3

4. 4

Q.13 Match List-I with List-II

Stages of Human Evolution Characteristics

- | | |
|----------------------|---------------------------------------|
| (A) Ramapithecus | (I) Lived in East African grasslands |
| (B) Australopithecus | (II) Lived near East and central Asia |
| (C) Homo habilis | (III) Hairy and walked like gorillas |
| (D) Neanderthal Man | (IV) First human like being |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.14 Ex situ conservation include:

1. Biosphere reserves
2. Zoological parks
3. National park
4. Sanctuaries

Options 1. 1

2. 2

3. 3

4. 4

Q.15 Arrange the given organs in correct sequence which are involved in life cycle of Plasmodium?

- (A) Gut of mosquito
- (B) Salivary glands of mosquito
- (C) Liver of man
- (D) Blood of man

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (D), (C), (A), (B)
3. (A), (D), (B), (C)
4. (B), (C), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.16 Which one of the following genotypes represents a similar skin color?

1. AABBCc and AaBBCC
2. AABbcc and AabBCc
3. AaBbcc and aaBBCC
4. aabbcc and AABBCc

Options 1. 1

2. 2
3. 3
4. 4

Q.17 Match List-I with List-II

List-I

List-II

Barriers

Examples

(A) Physical barrier

(I) Acid in the stomach

(B) Cellular barrier

(II) Interferons

(C) Cytokine barrier

(III) Leucocytes in our body

(D) Physiological barrier (IV) Skin on our body

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.18 The two alleles of a gene pair are located on -

1. Sex Chromosomes
2. Heterologous Chromosome
3. Homologous Chromosomes
4. Autosomes only

Options 1. 1

2. 2
3. 3
4. 4

Q.19 Insulin used in earlier times caused some patients to develop allergies because -

- (A). It was extracted from the pancreas of slaughtered cattle.
- (B). It was a foreign protein for the body of the patient.
- (C). It was a foreign lipid molecule for the body of the patient
- (D). It consists of two polynucleotide chains held together by disulphide bridges

Choose the correct answer from the options given below:

1. (A) and (C) only
2. (C) and (D) only
3. (B) and (C) only
4. (A) and (B) only

Options 1. 1

2. 2
3. 3

Q.20 Under Ganga Action and Yamuna Action plans, it is proposed to -

1. Directly discharge the water from industry into the rivers
2. Discharge untreated sewage into the Yamuna river
3. Discharge only the treated sewage into both the rivers
4. Discharge water with high BOD into the rivers

Options

1. 1
2. 2
3. 3
4. 4

Q.21 Which of the following is not a female contraceptive?

1. IUD
2. Cervical caps
3. Vasectomy
4. Pills

Options

1. 1
2. 2
3. 3
4. 4

Q.22 Which of the following factors can influence Hardy-Weinberg equilibrium?

- A. Natural selection**
B. Genetic recombination
C. Gene flow
D. Mutation

1. (A), (B) and (D) only
2. (A), (B), (C) and (D) only
3. (A), (C) and (D) only
4. (B) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.23 Which one of the following disease is not caused by bacteria?

1. Dysentery
2. Diphtheria
3. Common cold
4. Plague

Options

1. 1
2. 2
3. 3
4. 4

Q.24 The gynoecium represents the female reproductive part of the flower. When gynoecium is having more than one pistils and the pistils are fused together, the condition is called:

1. Syncarpous
2. Fused multicarpellary
3. Apocarpous
4. Apo-multicarpellary

Options

1. 1
2. 2
3. 3
4. 4

Q.25 Which one of the following cause of biodiversity loss is reflected in the phrase "need turns to greed"?

1. Habitat loss and fragmentation
2. Over-exploitation
3. Alien species invasion
4. Co-extinctions

Options 1. 1

2. 2
3. 3
4. 4

Q.26 Which of the following techniques are used for early diagnosis of diseases?

- (A). Urine analysis
- (B). PCR
- (C). ELISA
- (D). Recombinant DNA technology (RDT)

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.27 Arrange the given stages involved in replication of retrovirus.

- (A) New viral RNA is produced by the infected cell
- (B) Viral DNA is produced by reverse transcriptase
- (C) Viral DNA is incorporated into host genome
- (D) Viral RNA is introduced into animal cell

Choose the correct answer from the options given below:

1. (D), (B), (C), (A)
2. (A), (B), (C), (D)
3. (B), (A), (D), (C)
4. (D), (C), (B), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.28 Which one of the followings is responsible for producing the large holes in "Swiss cheese"?

1. *Saccharomyces cerevisiae*
2. *Propionibacterium sharmanii*
3. *Lactobacillus*
4. *Trichoderma polysporum*

Options 1. 1

2. 2
3. 3
4. 4

Q.29 Inactive protoxins is converted into its active toxin by:

1. Acidic pH of the gut
2. Basic pH of the gut
3. Neutral pH of saliva
4. Acidic pH of hemolymph

Options 1. 1

2. 2
3. 3
4. 4

Q.30 Which one of the following genetic law is responsible for 3:1 expression of the phenotypic ratio in F_1 generation?

1. Law of dominance
2. Law of segregation
3. Incomplete dominance
4. Co-dominance

Options 1. 1

2. 2
3. 3
4. 4

Q.31 Which one of the followings is an auto-immune disease?

1. Rheumatoid arthritis
2. Ascariasis
3. AIDS
4. Amoebic dysentery

Options 1. 1

2. 2
3. 3
4. 4

Q.32 Which one of the following hormones is essential for maintenance of the endometrium?

1. Progesterone
2. Follicle stimulating hormone
3. Estrogens
4. Oxytocin

Options 1. 1

2. 2
3. 3
4. 4

Q.33 Select the statements which are true for oogenesis?

- (A). Ovulation occurs at the secondary oocyte stage.
(B). Polar body is larger than the secondary oocyte.
(C). No more oogonia is formed after birth in the ovary.
(D). The sperm and the ovum nucleus fuse before the formation of the second polar body.

Choose the correct answer from the options given below:

1. (B) and (C) only
2. (A) and (B) only
3. (A), (C) and (D) only
4. (A) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.34 Pollen grains in majority of angiosperms are shed at:

1. 2-celled stage
2. 3-celled stage
3. 4-celled stage
4. 7-celled stage

Options 1. 1

2. 2
3. 3
4. 4

Q.35 Arrange the given stages of spermatogenesis in the correct sequence.

- (A). Spermatozoa
(B). Spermatocytes
(C). Spermatogonia
(D). Spermatids

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.36 Match List-I with List-II

List-I (Population interactions) List-II (Species A / Species B)

- | | |
|------------------|-------------|
| (A) Competition | (I) - / 0 |
| (B) Amensalism | (II) + / 0 |
| (C) Commensalism | (III) + / - |
| (D) Parasitism | (IV) - / - |

*** + ' sign for beneficial interaction, ' - ' for detrimental and ' 0 ' for neutral interaction**

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)
2. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
3. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.37 Which one of the following events does not occur during gene regulation in eukaryotes?

1. Replication level
2. Translational level
3. Transportation of mRNA from nucleus to cytoplasm
4. Transcriptional level

Options 1. 1

2. 2
3. 3
4. 4

Q.38 Which of the following structures are related to the male reproductive organ.

- (A). Isthmus
- (B). Sertoli cells
- (C). Leydig cells
- (D). Prostate

Choose the correct answer option given below.

1. (A), (B) and (C) only
2. (B), (C) and (D) only
3. (A), (B) and (D) only
4. (A) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.39 Arrange the given structures involved in packaging of DNA Helix.

- (A). Chromatin
- (B). Chromatin fibers
- (C). Nucleosome
- (D). Euchromatin

Choose the correct answer from the options given below:

1. (C), (A), (B), (D)
2. (A), (B), (C), (D)
3. (B), (A), (C), (D)

4. (A), (C), (D), (B)

Options 1. 1

2. 2
3. 3
4. 4

Q.40 Process of formation of new species due to change in allele frequency in a population is called:

1. Founder effect
2. Natural selection
3. Adaptive radiation
4. Industrial melanism

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Read the following passage carefully and answer the given questions

The separated DNA fragments can be visualized only after staining the DNA with ethidium bromide followed by exposure to UV radiation. Bright orange colored bands of DNA in ethidium bromide stained gel exposed to UV light can be seen. The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece. This step is known as elution. The DNA fragments purified in this way are used in constructing recombinant DNA by joining them with a cloning vector.

The extraction of separated bands of DNA from the gel piece is called:

1. Elution
2. Cloning
3. Transformation
4. Replication

Options 1. 1

2. 2
3. 3
4. 4

Q.42 Read the following passage carefully and answer the given questions

The separated DNA fragments can be visualized only after staining the DNA with ethidium bromide followed by exposure to UV radiation. Bright orange colored bands of DNA in ethidium bromide stained gel exposed to UV light can be seen. The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece. This step is known as elution. The DNA fragments purified in this way are used in constructing recombinant DNA by joining them with a cloning vector.

The natural polymer used in the agarose gel is extracted from:

1. Sea weeds
2. Desert plants
3. Soil microbes
4. Vegetable peels

Options 1. 1

2. 2
3. 3
4. 4

Q.43 Read the following passage carefully and answer the given questions

The separated DNA fragments can be visualized only after staining the DNA with ethidium bromide followed by exposure to UV radiation. Bright orange colored bands of DNA in ethidium bromide stained gel exposed to UV light can be seen. The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece. This step is known as elution. The DNA fragments purified in this way are used in constructing recombinant DNA by joining them with a cloning vector.

The DNA band appears orange due to –

1. Staining with acetocarmine and exposed to with uv light
2. Staining with ethidium bromide and exposed to uv light
3. Staining with ethidium bromide and exposed to infra red light

4. Staining with acetocarmine and exposed to infra-red light

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.44 Read the following passage carefully and answer the given questions

The separated DNA fragments can be visualized only after staining the DNA with ethidium bromide followed by exposure to UV radiation. Bright orange colored bands of DNA in ethidium bromide stained gel exposed to UV light can be seen. The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece. This step is known as elution. The DNA fragments purified in this way are used in constructing recombinant DNA by joining them with a cloning vector.

The DNA fragments are separated by-

- 1. Forcing them to move towards anode under an electric field
- 2. Forcing them to move towards cathode under an electric field
- 3. Forcing them to move towards anode under an electromagnetic field
- 4. Forcing them to move towards cathode under an magnetic field

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.45 Read the following passage carefully and answer the given questions

The separated DNA fragments can be visualized only after staining the DNA with ethidium bromide followed by exposure to UV radiation. Bright orange colored bands of DNA in ethidium bromide stained gel exposed to UV light can be seen. The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece. This step is known as elution. The DNA fragments purified in this way are used in constructing recombinant DNA by joining them with a cloning vector.

Which one of the following techniques is used to separate DNA fragments?

- 1. PCR
- 2. Autoradiography
- 3. Electron microscopy
- 4. Gel electrophoresis

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.46 Read the following passage carefully and answer the given questions

The earthworm is referred to as the farmer's friend. This is because they help in the breakdown of complex organic matter as well as in loosening of soil. Similarly, decomposers break down complex organic matter into inorganic substances like carbon dioxide, water and nutrients and the process is called decomposition. Dead plant remains such as leaves, bark, flowers and dead remains of animals, including fecal matter, constitute detritus, which is the raw material for decomposition. Some of the steps in the process of decomposition are fragmentation, catabolism, leaching, humification etc.

The process of breaking down of detritus into smaller particles is called:

- 1. Leaching
- 2. Catabolism
- 3. Fragmentation
- 4. Mineralisation

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.47 Read the following passage carefully and answer the given questions

The earthworm is referred to as the farmer's friend. This is because they help in the breakdown of complex organic matter as well as in loosening of soil. Similarly, decomposers break down complex organic matter into inorganic substances like carbon dioxide, water and nutrients and the process is called decomposition. Dead plant remains such as leaves, bark, flowers and dead remains of animals, including fecal matter, constitute detritus, which is the raw material for decomposition. Some of the steps in the process of decomposition are fragmentation, catabolism, leaching, humification etc.

Which one of the following is not associated with earthworm as 'farmer's friend'?

1. It reduces the quality of crops and soil
2. Faecal matter adds into fertility of soil
3. It helps with the breakdown of complex organic matter
4. It helps to loosen soil.

Options 1. 1

2. 2
3. 3
4. 4

Q.48 Read the following passage carefully and answer the given questions

The earthworm is referred to as the farmer's friend. This is because they help in the breakdown of complex organic matter as well as in loosening of soil. Similarly, decomposers break down complex organic matter into inorganic substances like carbon dioxide, water and nutrients and the process is called decomposition. Dead plant remains such as leaves, bark, flowers and dead remains of animals, including fecal matter, constitute detritus, which is the raw material for decomposition. Some of the steps in the process of decomposition are fragmentation, catabolism, leaching, humification etc.

Process of accumulation of dark coloured amorphous substance is called:

1. Mineralisation
2. Fragmentation
3. Leaching
4. Humification

Options 1. 1

2. 2
3. 3
4. 4

Q.49 Read the following passage carefully and answer the given questions

The earthworm is referred to as the farmer's friend. This is because they help in the breakdown of complex organic matter as well as in loosening of soil. Similarly, decomposers break down complex organic matter into inorganic substances like carbon dioxide, water and nutrients and the process is called decomposition. Dead plant remains such as leaves, bark, flowers and dead remains of animals, including fecal matter, constitute detritus, which is the raw material for decomposition. Some of the steps in the process of decomposition are fragmentation, catabolism, leaching, humification etc.

The degradation of detritus into simpler inorganic substances by the bacterial and fungal enzymes is called as-

1. Mineralization
2. Leaching
3. Fragmentation
4. Catabolism

Options 1. 1

2. 2
3. 3
4. 4

Q.50 Read the following passage carefully and answer the given questions

The earthworm is referred to as the farmer's friend. This is because they help in the breakdown of complex organic matter as well as in loosening of soil. Similarly, decomposers break down complex organic matter into inorganic substances like carbon dioxide, water and nutrients and the process is called decomposition. Dead plant remains such as leaves, bark,

flowers and dead remains of animals, including fecal matter, constitute detritus, which is the raw material for decomposition. Some of the steps in the process of decomposition are fragmentation, catabolism, leaching, humification etc.

The process of decomposition largely requires -

1. Carbon monoxide
2. Oxygen
3. Nitrogen
4. Sulphur dioxide

Options 1. 1

2. 2

3. 3

4. 4